

RESEARCH ARTICLE

Comparability and repeatability of three commonly used methods for measuring endurance capacity

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Abstract

Measures of endurance (time to exhaustion) have been used to address a wide range of questions in ecomorphological and physiological research, as well as being used as a proxy for survival and fitness. Swimming, stationary (circular) track running, and treadmill running are all commonly used methods for measuring endurance. Despite the use of these methods across a broad range of taxa, how comparable these methods are to one another, and whether they are biologically relevant, is rarely examined. We used Australian water dragons (*Intellagama lesueurii*), a species that is morphologically adept at climbing, swimming, and running, to compare these three methods of endurance and examined if there is repeatability within and between trial methods. We found that time to exhaustion was not highly repeatable within a method, suggesting that single measures or a mean time to exhaustion across trials are not appropriate. Furthermore, we compared mean maximal endurance times among the three methods, and found that the two running methods (i.e., stationary track and treadmill) were similar, but swimming was distinctly different, resulting in lower mean maximal endurance times. Finally, an individual's endurance rank was not repeatable across methods, suggesting that the three endurance trial methods are not providing similar information about an individual's performance capacity. Overall, these results highlight the need to carefully match a measure of performance capacity with the study species and the research questions being asked so that the methods being used are behaviorally, ecologically, and physiologically relevant.

KEYWORDS*Intellagama lesueurii*, lizard, locomotion, performance capacity, reptile, stamina, water dragon

1 | INTRODUCTION

Measures of endurance (time or distance until exhaustion during locomotion) have been widely used in ecology, morphology, and physiology research (e.g., Garland, 1994; Irschick & Higham 2016; Pough, 1978). As a metric, endurance capacity has provided insights into how diet and dehydration, foraging mode, habitat, temperature, and morphological and physiological trade-offs effect whole-organism performance (Bennett, 1980; Feder & Olsen, 1978; Huey, Bennett, John-Alder, & Nagy, 1984; Pinch & Claussen 2003; Vanhooydonck, Van Damme, & Aerts, 2001; Wilson & Havel 1989), as well as a proxy for survival and fitness (Garland, 1994; Irschick, Meyers, Husak, & Le Galliard, 2008; Lailvaux & Husak 2014; Le Galliard & Ferrières 2008). Research using measures of endurance are taxonomically diverse, and include amphibians (Herrel & Bonneaud 2012; Miller, 1982), birds (Ellerby, Cleary, Marsh, & Buchanan, 2003; Grubb, Jorgensen, & Conner, 1983), decapods (Full, 1987; Herreid & Full 1986), fishes (Videler & Wardle 1991), insects (Herreid & Full

1984; Lailvaux, Hathway, Pomfret, & Knell, 2005), and mammals (Buttemer, Story, Fildes, Baudinette, & Astheimer, 2008; Dohm, Hayes, & Garland, 1996; Matsumoto, Ishihara, Tanaka, Inoue, & Fushiki, 1996); however, reptiles, specifically lizards, have garnered the most attention in this field (e.g., Bennett, 1980; Garland, 1994; Irschick et al., 2008; Lacerda de Albuquerque, Bonine, & Garland, 2015; Pinch & Claussen 2003; Sorci, Swallow, Garland, & Clobert, 1995; Vanhooydonck et al., 2014).

Three frequently used methods for measuring endurance in lizards are running along either a stationary track (fixed running surface), a moving track (a treadmill), or swimming. Running on treadmills is the most commonly used method for measuring endurance (Garland, 1994; Lacerda de Albuquerque et al., 2015; Vanhooydonck et al., 2001) because the speed of the treadmill can be controlled and kept constant. Running on stationary tracks (Bennett, 1980; Clemente, Withers, & Thompson, 2009; Le Galliard & Ferrières 2008; Leal, 1999) and swimming trials (Feder & Olsen, 1978; Osborne, 2005; Uller & Olsson 2003) are alternatives to treadmills, and have the benefit of being lower in

cost, easily transportable, and do not require electricity to function (facilitating use in the field). Unfortunately, the ecological, morphological, or physiological rationale for selecting one method over another is often unclear, and brings into question whether the method being selected is biologically relevant for the given species or the research question (Careau & Garland 2012). To further complicate comparisons between studies, the methods used to measure an individual's endurance varies with respect to how they are calculated between studies, such as using the time until exhaustion from only a single trial (e.g., Clemente et al., 2009; Tobler, Healey, & Olsson, 2012), the mean time across several trials (e.g., Osborne, 2005), a maximum time across several trials (e.g., Garland, 1994; Lacerda de Albuquerque et al., 2015; Vanhooydonck et al., 2001), total distance moved (Bennett, 1980), or a calculated 0.9 quantile from several trials (Head, Hardin, & Adolph, 2012). Additionally, physiological mechanisms underpinning different endurance measures, such as maximum time to exhaustion (1 km/hr treadmill running) and maximum distance covered (circular stationary running track), are known to not be predicted by the same physiological factors in lizards (e.g., muscle oxidation capacity and anaerobic glycolic indicator enzymes, respectively; Garland, 1984). If the physiological drivers of performance are overlooked during endurance trial design, then the relevance of the information the tests are providing comes into question. Furthermore, there are physiological and biomechanical trade-offs with respect to performance for species adept at one mode of locomotion (e.g., climbing, flying, running, and swimming) having a decreased capacity to excel at another form of locomotion, particularly when they are mechanically conflicting (Lailvaux & Husak 2014; Vanhooydonck et al., 2001). This suggests that the information being provided by one endurance trial method may not be comparable with an alternative method, due to species-, population-, and individual-specific variation. Unfortunately, some methods are favored purely for logistical reasons (e.g., lab-based or fieldwork, equipment cost, prior experience with a given method, etc.), and in doing so, researchers may not be correctly addressing their questions or using species-appropriate tests. For example, most species of lizards can be induced to swim (Braun & Reif 1985); however, is the use of swimming as an endurance metric useful for a species that rarely, if ever, swims and which may recruit different muscles from those used in the terrestrial locomotion it is adapted for? Furthermore, swimming through water incurs a much higher level of drag compared to individuals running, potentially resulting in individuals tiring faster (Schmidt-Nielsen, 1972). As such, would this method be providing valuable or erroneous information? To address how comparable commonly used endurance trial methods are, comparison should be made that controls for species- and individual-specific variation to assess what type of information is being provided and how similar the results are.

To test the comparability and repeatability between three commonly used endurance trial methods (swimming in still water, running on a stationary track, and running on a treadmill), we examined how Australian water dragons (*Intellagama lesueurii*; a species adept at both running and swimming) perform within and across these methods. First, we examined if, within a method, repeated measures of an individual's endurance were consistent. We hypothesized that an individual lizard is always motivated to perform at its maximal capacity (i.e.,

providing a high level of performance repeatability), and expected that lizards would exhibit consistent performance across three repeated trials of each endurance trial method. Second, we tested if methods are comparable and if individuals perform at a consistent level across methods. If endurance trials were providing comparable information, then endurance would be similar among methods, and an individual's performance rank within the population would be consistent across method types.

2 | METHODS

2.1 | Study species and experimental design

Australian water dragons are large agamid lizards (maximum snout-vent length [SVL]: 304 mm; Thompson, 1993), with a wide distribution along the east coast of Australia (Cogger, 2014). These lizards are often found around bodies of freshwater associated with forested areas (Cogger, 2014), but can also be found in urban greenspaces and backyards (Littleford-Colquhoun, Clemente, Whiting, Ortiz-Barrientos, & Frère, 2017). Water dragons are fully adept at climbing, sprinting, and swimming (Baird, Lovern, & Shine, 2014; Cogger, 2014; Doody, Harlow et al. 2014), making this species an excellent candidate for comparing different endurance trial methods based on locomotion variation, for which they have specific morphological (Bellairs, 1970; Doody et al., 2014) and biomechanical adaptations (Ringma & Salisbury 2014).

We used yearling water dragons ($N = 45$) from our captive colony at Macquarie University. All lizards had previously been identically incubated as eggs (a constant 26.5°C; Harlow, 2001) and raised in common garden designed enclosures (semi-natural, 6.2 m² outdoor enclosures) and husbandry conditions (fed crickets three times weekly). We separated lizards into three test groups (A, B, and C, each $N = 15$) and tested them in a counter-balanced sequence to remove potential order effects from method type. All lizards were tested once per day (a single measure of endurance) and given a day's rest before the next trial. Each trial method was repeated three times, in the same sequence. Group A was first tested in the stationary track running trial (days 2, 8, and 14), followed by the swimming trial (days 4, 10, and 16) and then the treadmill trial (days 6, 12, and 18). Group B was tested on the same daily schedule in the order: swimming, treadmill, and stationary track running trial. Group C also tested on the same daily schedule in the order: treadmill, stationary track, and swimming. All trials and housing during the trial period (18 days; 1 acclimation day, 17 trial days) were within a temperature controlled room, set to their preferred body temperature of 30°C (Hosking, 2010). Before the trial period, we measured each individual's initial mass (± 0.01 g) and SVL (± 1 mm), which allowed us to calculate initial body condition (see *Statistical Analysis*). Before each individual trial, we recorded their body temperature to the nearest 0.5°C with a handheld laser thermometer (Berg, Theisinger, & Dausmann, 2015; Bucklin, 2006) held to the lizard's abdomen (Carretero, 2012) so that body temperature could be controlled for between and across individuals and methods. Lizards were motivated to swim or run by gently chasing them using a gloved hand (blue nitril glove; similar to that of Carazo, Noble, Chandrasoma, & Whiting, 2014;

Riley, Noble, Byrne, & Whiting, 2017). As the lizard began to tire, the base of its tail was gently pinched and lifted, spurring it to continue to move. The time (sec) from when a lizard began a trial to when it was considered exhausted was used as a metric for the individual's endurance.

2.2 | Endurance trial setup

2.2.1 | Swimming circuit

Swimming trials were conducted in a 60 L round plastic tub with a weighted 20 L round plastic bucket placed in the center. The placement of the bucket within the tub created a 1.2 m circuit between the inside wall of the tub and the outer wall of the bucket (0.08 m wide). The tub was filled with 10 cm of 30°C water (matching the air temperature of the testing room); deep enough to prevent the lizard from touching the bottom and ensuring the muscles being used for locomotion were those purely for swimming, rather than running through shallow water. Lizards were considered exhausted when they closed their eyes, began to list in the water, and ceased swimming after having the base of their tail pinched ten times (Garland & Else 1987).

2.2.2 | Stationary track

The stationary track running trials were conducted with an identical setup as the swimming track (a 60 L round plastic tub with a 20 L round plastic bucket fixed in the center). The substrate had a light dusting of soil (5 mm) to provide traction on the smooth plastic base of the tub (Bennett, 1980; Huey et al., 1984; Qi, Noble, Wu, & Whiting, 2014), preventing individuals from losing traction or slipping as they ran. Lizards were considered exhausted when they lowered their body and head to the ground, and no longer ran after having the base of their tail pinched ten times (Garland & Else 1987).

2.2.3 | Treadmill track

The treadmill running trials were conducted on a human treadmill set to a fixed running speed of 1.0 km/hr (Garland & Else 1987; Noble, Fanson, & Whiting, 2014). A 1.1 m long clear plastic box with an adjustable central compartment was placed over the treadmill, with the middle compartment adjusted to the same width as both the stationary running and swimming track (0.08 m). This adjustable compartment prevented the lizard from turning around during a trial. During the trial lizards were placed in the first third (0.36 m) of the treadmill, so that as they tired there would be enough time to pinch the base of the tail 10 consecutive times (Garland & Else 1987) before the lizard was carried along the track and pushed off the end.

2.3 | Statistical analyses

All statistical analyses were performed in R (version 3.2.3; R Core Team 2015). Prior to conducting analyses, we explored the distribution of the data following Zuur, Ieno, and Elphick (2010), and we did not find any unexplainable outliers and there was no strong collinearity between predictor variables.

2.3.1 | Within-method comparison

First, we examined how individual endurance differed across three repeated measurements within each endurance trial method separately (running, swimming, and treadmill). We subsetted our data for each endurance trial method, and performed a linear mixed-effects model (LMM) using the function `lme` in the R package `nlme` (Pinheiro, Bates, DebRoy, & Sarkar, 2015) that examined how endurance was affected by the fixed effects of order, lizard body condition (residuals from simple linear regression between log SVL and log mass; Cox, Duryea, Najarro, & Calsbeek, 2011), and lizard body temperature. We also controlled for dependencies in our data due to repeated measurements of the same lizard (lizard identity), individuals from the same clutch (mother identity), and individuals housed in the same enclosure. We performed this model three times in total, once for each method (e.g., running endurance, swimming endurance, and treadmill endurance).

We then separately calculated adjusted repeatability (R_{adj}) in endurance while controlling for the same fixed and random effects as were in our LMM. Repeatability (R) measures were used to quantify if lizard endurance was consistent within and between each measure of endurance. Adjusted repeatability (R_{adj} ; a type of repeatability measure that controls for potential confounding effects; Nakagawa & Schielzeth 2010) and 95% confidence intervals were calculated using the `rpt` function from the `rptR` R package using 1000 permutations (Schielzeth & Nakagawa 2013).

We used the results from these LMMs (see *Results*), to guide the analysis for our subsequent research goals: to compare endurance across the three trial methods. In order to have a metric that was comparable across all methods, we used maximal endurance because (1) we wanted our analysis to reflect the techniques that are commonly used in endurance research (Garland, 1994; Irschick et al., 2008; Lacerda de Albuquerque et al., 2015; Vanhooydonck et al., 2001), (2) we observed opposing order effects in repeated measures of endurance within the running and treadmill methods, and (3) we observed low repeatability of individual endurance times across all methods so the maximal endurance time was likely to be a true estimate (see *Results*). Using maximal endurance as our metric allowed a straight-forward comparison of endurance both among the same individual across methods, as well as differences in endurance among methods for all sampled individuals (at the population level).

2.3.2 | Between-method comparison

To compare maximum endurance between methods, we performed a LMM using the function `lme` in the R package `nlme` (Pinheiro et al., 2015). The model had maximum endurance as the response variable, and included the fixed factors of method (categorical: swim, run, or treadmill), order the test was presented to the lizard (continuous), body condition (continuous), and body temperature (continuous). It also included the random factors of lizard identity to control for dependency among repeated observations of the same lizard ($N = 45$), mother identity to control for dependency among observations of related individuals ($N = 34$), and housing tub to control for dependency among observations of individual within the same rearing environment

($N = 11$). We used the function `glht` from the R package `multcomp` (Hothorn, Bretz, & Westfall, 2008) to calculate Tukey contrasts between our three endurance trial methods, and statistically assess differences. Visual inspection of the residual plot for this model did not reveal any obvious deviations from homoscedasticity or normality of residuals. Parameter estimates were considered significant if $P < 0.05$, and when we predicted data for visualizations from the models, we set the continuous fixed factors to the mean and the factorial fixed factors to the intercept-level values.

Then, to assess repeatability between methods, we formed a new dataset with only the maximum endurance each individual lizard performed across our three replicates (e.g., only one measure of endurance per method per lizard). We also wanted to ensure each method was comparable, which once we compared it statistically (see below for *Methods and Results*), we found that they were not. So, we transformed each method's endurance data to rank order (Irschick, Herrel, Vanhooydonck, Huyghe, & Van Damme, 2005) to assess if an individual's rank was consistent across methods by calculating R_{adj} as above.

3 | RESULTS

3.1 | Within-method comparison

Neither body condition nor temperature significantly affected running endurance (sec), which decreased significantly across our repeated measures (i.e., a negative order effect; Table 1A). Swimming endurance (sec) was not affected by order, body condition, or temperature (Table 1B). Treadmill endurance (sec) increased significantly across our repeated measures (i.e., a positive order effect; Table 1C) and there was a positive relationship between treadmill endurance and body temperature (Table 1C). Treadmill endurance was not affected by body condition (Table 1C).

R_{adj} across replicates of running, swimming, and treadmill trials was 0 (95% CI = 0, 0.16), 0.21 (95% CI = 0, 0.44), and 0.37 (95% CI = 0, 0.54), respectively. Because all of the confidence intervals overlap 0, the measures of endurance provided by these methods are no more repeatable than expected by chance alone.

3.2 | Between-method comparison

Mean maximal endurance (sec) measured on a treadmill and a stationary track were not significantly different ($\beta = -2.14$, $z = -0.48$, $P = 0.88$). Time to exhaustion (sec) while swimming was significantly lower than running endurance ($\beta = -22.04$, $z = -5.13$, $P < 0.01$) and treadmill endurance ($\beta = -19.90$, $z = -4.89$, $P < 0.01$; Figure 1). Endurance was significantly negatively related to lizard body condition, but not any of our other model factors (Table 2). With respect to generalized performance information, lizard rank order was not consistent between methods. R_{adj} was 0.08 (95% CI = 0, 0.25) and this reflects low consistency that is no more repeatable than one would expect from chance alone.

TABLE 1 Results from Linear Mixed Effects Models Examining if Lizard (A) Running, (B) Swimming, and (C) Treadmill Endurance is Affected by Order of Repeated Measures, Body Condition, or Body Temperature

(A) Running ($N_{\text{obs}} = 135$; $N_{\text{ind}} = 45$; $N_{\text{mom}} = 34$; $N_{\text{tub}} = 11$)				
Fixed Effects	β	SE	t Value	P
Intercept	97.67	44.41	2.20	0.03*
Order	-6.94	2.45	-2.84	0.01*
Body condition	-0.66	0.81	-0.81	0.42
Body temperature	-0.46	1.53	-0.30	0.77
Random Effects	σ^2			
Individual	0.00004			
Clutch	0.00004			
Tub	0.16			
Residual	0.05			
Repeatability	R_{adj}	Lower CI	Upper CI	
	0	0	0.16	

(B) Swimming ($N_{\text{obs}} = 135$; $N_{\text{ind}} = 45$; $N_{\text{mom}} = 34$; $N_{\text{tub}} = 12$)				
Fixed Effects	β	SE	t Value	P
Intercept	219.38	44.83	4.89	0.00*
Order	5.92	2.09	2.83	0.006*
Body condition	-2.09	1.14	-1.84	0.08
Body temperature	-5.55	1.49	-3.72	0.0004*
Random Effects	σ^2			
Individual	0.01			
Clutch	0.01			
Tub	0.0001			
Residual	0.04			
Repeatability	R_{adj}	Lower CI	Upper CI	
	0.37	0	0.54	

(C) Treadmill ($N_{\text{obs}} = 135$; $N_{\text{ind}} = 45$; $N_{\text{mom}} = 34$; $N_{\text{tub}} = 11$)				
Fixed Effects	β	SE	t Value	P
Intercept	26.30	30.60	0.86	0.39*
Order	-0.47	1.39	-0.34	0.73
Body condition	-0.55	0.65	-0.85	0.40
Body temperature	-1.04	1.02	1.01	0.31
Random Effects	σ^2			
Individual	0.35			
Clutch	0.35			
Tub	0.000002			
Residual	0.02			
Repeatability	R_{adj}	Lower CI	Upper CI	
	0.21	0	0.44	

Significant effects are indicated with an asterisk (*) to the right of the P value. Our models also included the random effects of lizard identity, mother identity, and housing tub and the variance for each of these random effects are presented. Adjusted repeatability (R_{adj}) of endurance time, and the corresponding 95% confidence intervals, are also presented.

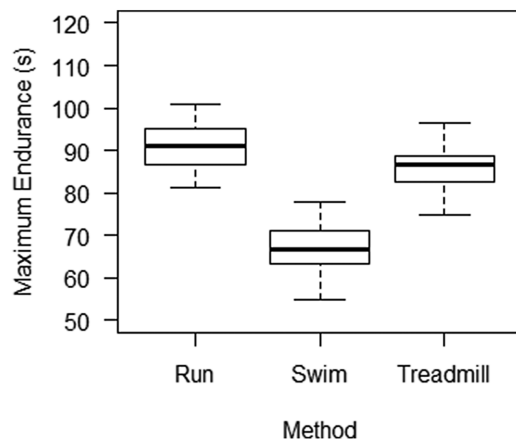


FIGURE 1 A comparison of maximum endurance (sec) between our three endurance trial methods (Run, stationary running track; Swim, swimming circuit; Treadmill, treadmill running track) using predicted data. Mean maximum endurance (sec) was significantly lower in swimming trials than running and treadmill trials (as indicated from different letters between boxplots)

TABLE 2 Linear Mixed Effects Model ($N_{\text{obs}} = 405$, $N_{\text{ind}} = 45$, $N_{\text{mom}} = 34$, $N_{\text{tub}} = 11$) Examining if Lizard Endurance Differs Between Methods (Running, Swimming, or Treadmill Trials)

Fixed Effects	β	SE	t Value	P
Intercept (Run)	135.01	39.69	3.40	<0.01*
Method: Swim	-22.04	4.29	-5.13	<0.01*
Method: Treadmill	-2.14	4.43	-0.48	0.63
Body condition	-1.47	0.72	-2.03	0.05*
Order	-1.11	0.71	-1.56	0.12
Body temperature	-1.38	1.37	-1.01	0.32

Our model also included the fixed effects of lizard body condition, order of testing, and body temperature. Also, it included the random effects of lizard identity, mother identity, and housing tub. Significant effects are indicated with an asterisk (*) to the right of the p-value. Multiple comparisons between all methods using Tukey HSD contrasts are shown in our Results section.

4 | DISCUSSION

Endurance was not repeatable across three trials within any of the three methods (within-method repeatability). This finding does not support our expectation that individuals would always be motivated to perform at their maximal capacity within an endurance trial method. Between methods, water dragon maximal endurance was on average significantly lower when swimming than in the two running methods (treadmill and stationary track), whereas mean maximal endurance in the two running methods did not significantly differ from each other. This finding demonstrates that endurance trial methods using similar locomotion modes are comparable with respect to mean maximal endurance. Finally, we found that rank order endurance was not repeatable across the three methods (between-method repeatability); for example, individual lizards exhibiting relatively high endurance on the treadmill would not necessarily have relatively high endurance in swimming or stationary track trials. This finding contradicts a com-

mon convention that metrics of endurance capacity, in this case time to exhaustion, are interchangeable.

In our investigation of repeated measurements of endurance within each trial method type (running, swimming, and treadmill), we interestingly found order effects within two of the three. Order negatively affected stationary running track endurance, with individuals having lower endurance times during the last trial than the first, suggesting that they may become habituated to the motivation (Husak, Keith & Wittry, 2015). Conversely, we saw positive order effects within the treadmill endurance trial methods, with individuals increasing endurance times from the first to last trial, suggesting they were improving at the activity (i.e., increased performance through training; Husak et al. 2015). Within the swimming endurance trial method, we did not detect any order effects. This result underscores the difference in how a single species of lizards can respond with respect to repeated measures based on method type. Similar antithetical results in order effects have previously been reported between different performance metrics (i.e., sprinting and endurance running) in green anoles (*Anolis carolinensis*; Husak et al. 2015). However, endurance measures in another Australian agamid lizard (*Ctenophorus nuchalis*), did not exhibit order effects, even under experimental training regimes, yet substantial pathological difference were seen between treatment groups (e.g., difference in cardiac and skeletal muscle size and enzymatic activity; Garland, Else, Hulbert, & Tap, 1987). Our findings demonstrate how performance varies not only between methods but also over time. Thus, timing of repeated measures within a study should be given attention. If one uses maximum endurance time, it may require longer delays between repeated measure to prevent habituation, or “training-effects.” Alternatively, one could use mixed-models, as we did in this study, to investigate and control for potential confounding order effects (also see Careau & Wilson 2017). Through the use of LMMs, studies are able to control for the presence potential order-effects (e.g., test habituation or training-effects) and thus analysis data without the confounding effect of increased/decreased performance over time that has pervious proved challenging in performance literature. A further benefit to the use of mixed models is that can be used to investigate ecological trade-offs, and behavioral plasticity in performance metrics (Careau & Wilson 2017). Overall, we strongly suggest not relying on a single measure of endurance, or a pooled mean of repeated measures as these are likely to produce spurious results, and to ensure the metrics and methods being used are ecologically and biologically relevant for the study species and research question.

Regardless of these order effects, individual repeatability within methods was extremely low. This means that lizards were not always performing at their maximal capacity. Previous lizard performance studies have reported similar findings, and have demonstrated that sub-maximal performance can lead to skewed results (Losos, Creer, & Schulte, 2002). However, consistent repeatability of endurance measures has been observed across numerous taxa, including horses (Tolley, Notter, & Marlowe, 1983), lizards (Huey, Dunham, Overall, & Newman, 1990; Van Berkum, Huey, Tsuji, & Garland, 1989; Uller & Olsson 2003), salamanders (Austin & Shaffer, 1992; Shaffer, Austin, & Huey, 1991), and snakes (Jayne & Bennett, 1990a). Yet, the duration these measures remain repeatable can dramatically vary. For example,

tiger salamander (*Ambystoma californiense*) endurance is repeatable over a period on multiple days but not over multiple months (Austin & Shaffer, 1992), whereas the endurance measure of neonate fence lizards (*Sceloporus occidentalis*) could predict their performance over a year later (Van Berkum et al., 1989). This suggests that although repeatable long-term endurance measures are achievable, caution is required to ensure that the trial type being used (e.g., swimming, continuous treadmill running, or stationary track running), and the motivating stimulus being used, is species appropriate and effective. A potential source of our low repeatability was that our motivating stimulus (a gloved hand) was not eliciting maximal performance. We suggest that researchers should test for individual repeatability of endurance measurements for each species and study to inform their analysis on a case-by-case basis through pilot studies.

One argument made for the interchangeability of endurance trial methods is that the general information being provided would similarly trend between methods (e.g., distance covered and time to exhaustion; Jayne & Bennett 1990b). For example, an individual with a relatively high endurance score running on a treadmill should also have a relatively high score during a running trial on a stationary track or during a swimming trial. However, when our individual maximum endurance times for the water dragons were ranked for each endurance trial method, we found that the ranks of individuals were not consistent between methods. This suggests that general trends observed from one method would not necessarily be the same as another, and if this information was to be used as a proxy for fitness or survivorship it may be misleading. This finding was similar to previous comparisons using field- and lab-based performance trials in lizards that demonstrated that differences between age, behavior, motivation, species, and substrate use, make comparing different performance measures difficult (Irschick, 2003; Irschick et al., 2005; Mcelroy, Meyers, Reilly, & Irschick, 2007). Due to the experimental design of our study (i.e., lizards raised in a common-garden setting), we were able to control for some predictors of performance including age (Van Berkum et al., 1989; Vanhooydonck, Herrel, Van Damme, Meyers, & Irschick, 2005), incubation environment (Van Berkum et al., 1989; Van Damme et al. 1992), diet (Husak et al. 2015), and habitat structure (Kohlshorff & Navas 2007). Even with these performance predictors controlled for, lack of repeatability between methods were not overly surprising if one considers it within the context of an individual's variation in behavior and physiology (Careau & Garland 2012; Lailvaux & Husak 2014). Individual variation in enzyme activity, blood chemistry, metabolism, and aerobic and anaerobic capacity can influence endurance capacity (Garland 1984; Careau & Garland 2012; Norin & Malte 2012). As such, the high variability in overall performance within and between our trial methods is likely driven by behavioral, morphological, and physiological responses to different trial methods (Garland, 1994; Losos et al., 2002; Careau & Garland 2012).

In comparing our three endurance trial methods, we found that the swimming method led to exhaustion significantly faster than the two running methods (i.e., treadmill and stationary track), and that stationary track and treadmill running resulted in mean maximal endurance that was not significantly different from one another. This was not overly surprising for two reasons: (1) mechanically and physiologically,

swimming and running are very different activities in lizards, with different muscle groups being recruited (Bellairs, 1970; Garland, 1984; Ringma & Salisbury 2014); and (2) moving through water may result in individual tiring sooner compared to moving through air (Schmidt-Nielsen, 1972). Running is driven primarily by the hind legs, with the muscle of the thighs providing much of the propulsion, whereas a side-to-side bending of the body by axial muscles is used to increase stride length (Bellairs, 1970; Ritter, 1996). Conversely, swimming in semi-aquatic lizards, such as water dragons, is driven primarily by their laterally compressed tail acting as a paddle with side-to-side bending of the body by axial muscles provide propulsion and the limbs tucked against the body (Bellairs, 1970; Ringma & Salisbury 2014; Ritter, 1996). Furthermore, the medium with which an animal is moving through during a swimming trial (water) is substantially denser than that of a running trial (air), resulting in much higher drag for individuals when swimming compared to when running (Schmidt-Nielsen, 1972). Logically, this would tire lizards sooner resulting in lower swimming endurance times compared with running trials. This only further compounds differences between aquatic and terrestrial locomotion, and underscores the importance of ensuring the biological relevance of a given trial type (Bennett & Huey, 1990). In general, although the two running trials may provide comparable results, it is inadvisable to directly compare endurance trial methods that use different forms of locomotion.

In summary, our study underscores the importance of considering the functional relevance of a specific endurance trial method for a given species, as well as the context in which the question of individual endurance is being asked (Bennett & Huey, 1990). For example, does the study species engage in long-distance runs? If so, a slow continuous treadmill may be most appropriate. Or, does it typically make a series of short bursts? If so, a stationary track may be more appropriate. Furthermore, it is important to include population as a factor, as there may be variation in locomotor ability that is population-specific (Careau & Garland 2012; Garland, 1994). In general, we advise against contradictions, such as swimming as a measure of endurance in an arid-adapted, terrestrial lizard. Instead, we suggest selecting an endurance method that complements the research question, and how the specific behavior, ecology, and physiology of the study species suit the intended method. The potential for measures of endurance to provide valuable information spans well beyond morphological and physiological research (Irschick et al., 2008; Vanhooydonck et al., 2001), providing insight for conservation and wildlife management (Adams, Hoover, & Killgore, 2000), invasive ecology (Llewellyn, Phillips, Alford, Schwarzkopf, & Shine, 2010), and behavioral ecology (Osbourne, 2005). By simply ensuring that ecologically relevant methods are applied, we can go a long way to understanding the importance of physiological performance capacity to reproductive fitness and evolutionary ecology.

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CONFLICT OF INTEREST

None.

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